

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.2 LINEAR MOTION: FINAL EXPLANATION – THE MKOKOTENI MYSTERY SOLVED

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation. Think back to the runaway mkokoteni loaded with sacks of maize on the Limuru–Nairobi escarpment road. Answer every question in the space under it, working on your own. For calculations, write GIVEN, FIND, the equation you choose, your working and your answer with units. Use $g = 10 \text{ m/s}^2$ for free fall. Use the scientific words you learned (distance, displacement, speed, velocity, acceleration). Write in full sentences when you are asked to explain.

Part 1: Why does the mkokoteni keep getting faster?

The mkokoteni was released from rest at the top of the Limuru escarpment. Nobody pushed it and it has no engine, yet witnesses said it became dangerously fast within seconds.

(a) Some people in your Lesson 1 discussion said 'something must be pushing it'. Use your knowledge of gravity to explain what is really happening to the cart's motion.

(b) What do we call the continuous change in velocity of the cart? Explain how it is different from speed.

(c) Explain why the cart's speed keeps increasing the whole time it rolls downhill, rather than increasing once and then staying the same.

(a) No person is pushing the cart. Gravity is an invisible force that acts on the cart all the time. Because the road slopes downhill, gravity gives an unbalanced force down the slope. This force makes the cart start moving from rest and keeps making it move faster.

(b) The continuous change in velocity is called acceleration. Speed only tells us how fast the cart is moving (how much distance it covers each second). Acceleration tells us how quickly its velocity changes each second, so a cart that speeds up has acceleration even though nobody is pushing it.

(c) Gravity does not stop acting while the cart rolls. The unbalanced force down the slope is there at every moment, so the cart keeps gaining velocity every second. This produces a constant acceleration, and the speed grows steadily with time instead of levelling off. The cart starts slowly because it begins from rest, but it becomes dangerously fast within seconds because its speed adds up every second.

Part 2: Predicting speed, distance and time with the equations

Your class found that the mkokoteni's constant acceleration down the escarpment is about 3 m/s^2 . It starts from rest ($u = 0$).

(a) Which equations of linear motion would you choose to find its speed and its distance after 4 s? Write the GIVEN and FIND quantities first, then calculate both.

(b) Witnesses said the cart was moving at about 40 m/s when it crashed. How long had it been rolling at that moment? Show your working.

(c) Use $v^2 = u^2 + 2as$ to find how far down the road the cart had travelled when it reached 40 m/s. Give your answer to the nearest metre.

(d) Explain, using your answers, why we can say we CAN predict how fast and how far the cart travels before it crashes.

(a) GIVEN: $u = 0$, $a = 3 \text{ m/s}^2$, $t = 4 \text{ s}$. FIND: v and s . For speed use $v = u + at = 0 + (3)(4) = 12 \text{ m/s}$. For distance use $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(3)(4^2) = \frac{1}{2}(3)(16) = 24 \text{ m}$. After 4 s the cart is moving at 12 m/s and has travelled 24 m.

(b) GIVEN: $u = 0$, $v = 40 \text{ m/s}$, $a = 3 \text{ m/s}^2$. FIND: t . Use $v = u + at$, so $t = (v - u)/a = (40 - 0)/3 \approx 13.3 \text{ s}$. The cart had been rolling for about 13 seconds.

(c) GIVEN: $u = 0$, $v = 40 \text{ m/s}$, $a = 3 \text{ m/s}^2$. FIND: s . Use $v^2 = u^2 + 2as$, so $40^2 = 0 + 2(3)s$, $1600 = 6s$, $s = 1600/6 \approx 267 \text{ m}$. The cart had travelled about 267 m (roughly 270 m) down the escarpment.

(d) The acceleration is constant, so the equations of linear motion give exact numbers for speed, distance and time at any moment. Once we know a and that $u = 0$, we can work out how fast the cart is going and how far it has rolled. For example, it is already at 12 m/s after 4 s, and it reaches 40 m/s (144 km/h) after about 267 m. So the speed and the crash distance are predictable, not a mystery.

Part 3: Speed along the escarpment and a model of the cart

Use $a = 3 \text{ m/s}^2$ and $u = 0$ for the mkokoteni. Complete the table by calculating the speed v at each distance marker from the start, using the correct equation.

Distance from start, s (m)	Speed, v (m/s)
0	
50	

(a) Use $v^2 = u^2 + 2as$ with $u = 0$, so $v = \sqrt{2as} = \sqrt{6s}$.

Distance from start, s (m)	Speed, v (m/s)
0	0
50	17.3
100	24.5
200	34.6

Sample working at 50 m: $v^2 = 0 + 2(3)(50) = 300$, so $v = \sqrt{300} \approx 17.3 \text{ m/s}$.

(b) No. The speed goes from 17.3 m/s to 24.5 m/s, which is an increase of about 1.4 times, not 2 times. Speed is not directly

100	
200	

(a) Fill in the speed column (one decimal place) and show one full calculation.

(b) From 50 m to 100 m the distance doubles. Does the speed double as well? Use your table to answer and explain what this tells you about the relationship between speed and distance.

(c) You are drawing the class wall model of the mkokoteni descending the escarpment. Describe what the velocity arrows at these four markers should look like, and name two more labels or symbols you would add to the model to show how a crash could be prevented.

proportional to distance. The square of the speed is proportional to distance ($v^2 = 2as$), so doubling the distance multiplies v^2 by 2 but multiplies v by only about 1.4.

(c) The arrows should start with no arrow (or a zero-length arrow) at 0 m and get longer at 50 m, 100 m and 200 m in proportion to the speeds in the table. The longest arrow, at 200 m, should be about twice as long as the one at 50 m. Other labels I would add: the symbol g or an arrow for the downward pull of gravity, the symbols u , v , a , s and t , and safety mitigation arrows such as a crash barrier or a warning sign placed at a calculated distance along the road before the cart reaches a dangerous speed.

Part 4: Free fall – the mango and the cart

Your class dropped a rubber bung, a crumpled paper ball and a small stone from 2 m and observed what happened. Now think about a ripe mango falling from a tree near Kisumu and the mkokoteni on the escarpment. Use $g = 10 \text{ m/s}^2$.

(a) Describe what you saw in the drop experiment when two objects of different mass were dropped together. Explain why a heavy object does not fall faster than a light one.

(a) When two objects of different mass were dropped together from 2 m, they landed at the same time. A heavier object has a greater gravitational pull (weight, $W = mg$), but it also has more mass to accelerate. The two effects cancel ($a = F/m = mg/m = g$), so every freely falling object has the same acceleration, about 10 m/s^2 (9.8 m/s^2 in our experiment), whatever its mass.

(b) GIVEN: $u = 0$, $a = g = 10 \text{ m/s}^2$, $t = 2 \text{ s}$. Speed: $v = u + at = 0 + (10)(2) = 20 \text{ m/s}$. Distance: $s = \frac{1}{2}at^2 = \frac{1}{2}(10)(2^2) = 20 \text{ m}$. The mango hits the ground at 20 m/s after falling 20 m .

(c) Free fall is the limiting case of the same gravitational acceleration, when the object falls straight down with no slope and no friction. The mkokoteni is on a slope, so only part of gravity's pull acts along the road, and friction also acts against it. That is why its acceleration (about 3 m/s^2) is less than g . Both motions are constant acceleration caused by gravity, so the same equations of linear motion describe both.

(b) A mango is released from rest and falls freely for 2 s. Calculate its speed after 2 s and the distance it falls.

(c) The mkokoteni's acceleration (about 3 m/s^2) is smaller than g . Explain how free fall is related to the mkokoteni's motion, and say why the mkokoteni's acceleration is not equal to g .

Part 5: Road safety and the Pitch to the County Engineer

In Lesson 6 you studied a matatu braking on Mombasa Road. Suppose the matatu brakes with a constant deceleration of 5 m/s^2 until it stops ($v = 0$). Use $v^2 = u^2 + 2as$ with $a = -5 \text{ m/s}^2$. Convert km/h to m/s by dividing by 3.6.

(a) Calculate the braking distance for a matatu travelling at 30 km/h and at 90 km/h (give your answers to one decimal place).

(b) Compare the two distances. What happens to braking distance when speed is tripled, and what does this tell you about the relationship between speed and braking distance?

(c) Write a short pitch (4–6 sentences) to the Limuru County Engineer. Explain why the mkokoteni kept accelerating, what mathematics could have predicted the crash, and one evidence-based safety measure you recommend for the escarpment road. In your answer, answer the KICD questions: how is free fall important in real life, and how

(a) At 30 km/h: $u = 30 \div 3.6 \approx 8.33 \text{ m/s}$. $0 = (8.33)^2 + 2(-5)s$, so $s = 69.4/10 \approx 6.9 \text{ m}$. At 90 km/h: $u = 90 \div 3.6 = 25 \text{ m/s}$. $0 = 25^2 + 2(-5)s$, so $s = 625/10 = 62.5 \text{ m}$.

(b) $62.5 \div 6.9 \approx 9$, so the braking distance is about nine times longer when the speed is three times greater. Braking distance is proportional to the square of the initial speed ($s = u^2/2a$), so it is not a straight-line relationship. A small increase in speed causes a much larger increase in stopping distance. This is why speeding on steep roads is so dangerous.

(c) Dear County Engineer, the mkokoteni kept getting faster because gravity is an unbalanced force that pulls it down the slope all the time. This gives it a constant acceleration of about 3 m/s^2 , so its speed grows every second without any engine or push. Using $v = u + at$ and $v^2 = u^2 + 2as$, we can calculate that the cart reaches about 40 m/s (144 km/h) after about 13 s and about 267 m, so the crash could have been predicted. I recommend crash barriers or runaway ramps at distances calculated from these equations, together with warning signs and speed limits on the escarpment. Free fall matters in real life because the same equations help us plan safe construction sites, design roads and understand falling objects such as mangoes. Its dangers can be reduced by keeping a safe distance, controlling speed, using barriers and wearing protective equipment, so that mathematics helps us prevent crashes.

can its dangers be reduced?	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific explanation of the phenomenon	Clearly explains that gravity is a constant unbalanced force down the slope that produces constant acceleration, rejects the 'something is pushing it' idea, and links acceleration to velocity changing every second.	Identifies gravity as the cause of the increasing speed and names acceleration, but the explanation of why it is continuous or constant is incomplete.	Gives vague or incorrect reasons (for example, a push or the cart's mass alone) or confuses speed with acceleration.
Use of equations of linear motion	Lists GIVEN and FIND, selects the correct equation every time, rearranges correctly and gets correct answers with units (12 m/s, 24 m, 13.3 s, 267 m).	Selects mostly correct equations and gets most answers right, with small slips in rearranging, rounding or units.	Chooses incorrect equations, omits working or units, or cannot substitute values correctly.
Interpreting data and relationships	Completes the table accurately and explains the non-linear relationships (speed against distance, braking distance against speed) using the equations and the 1.4 times and nine times comparisons.	Completes the table and recognises that the relationships are not straight-line, but gives limited supporting numerical evidence.	Table has several errors or the student assumes that doubling distance or tripling speed simply doubles or triples the outcome.
Free fall and the mass question	Uses observations from the drop experiment, explains with $a = F/m = g$ why mass does not matter, and links free fall to the mkokoteni as a case of the same gravitational acceleration.	States that all objects fall with the same acceleration and calculates free-fall values correctly, but the explanation of why mass does not matter is brief.	Says heavier objects fall faster or gives no reason for equal acceleration.
Real-life application and communication	The pitch is clear and well organised, uses Kenyan road-safety evidence and calculated values, gives a sensible mitigation measure, and answers both KICD questions using correct scientific vocabulary.	The pitch covers the explanation and one safety measure, with some use of numbers and vocabulary.	The pitch is unclear or incomplete, gives no safety measure, or does not link the mathematics to the real situation.

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